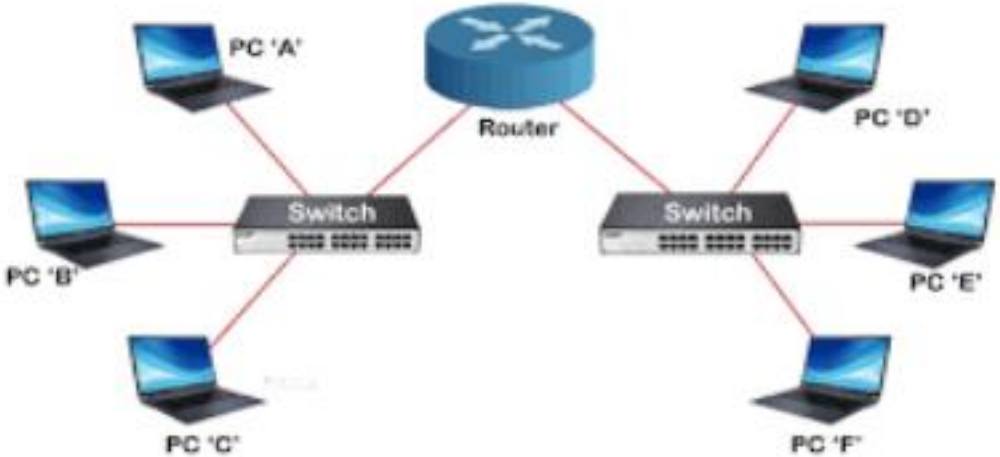
Networking concepts and security

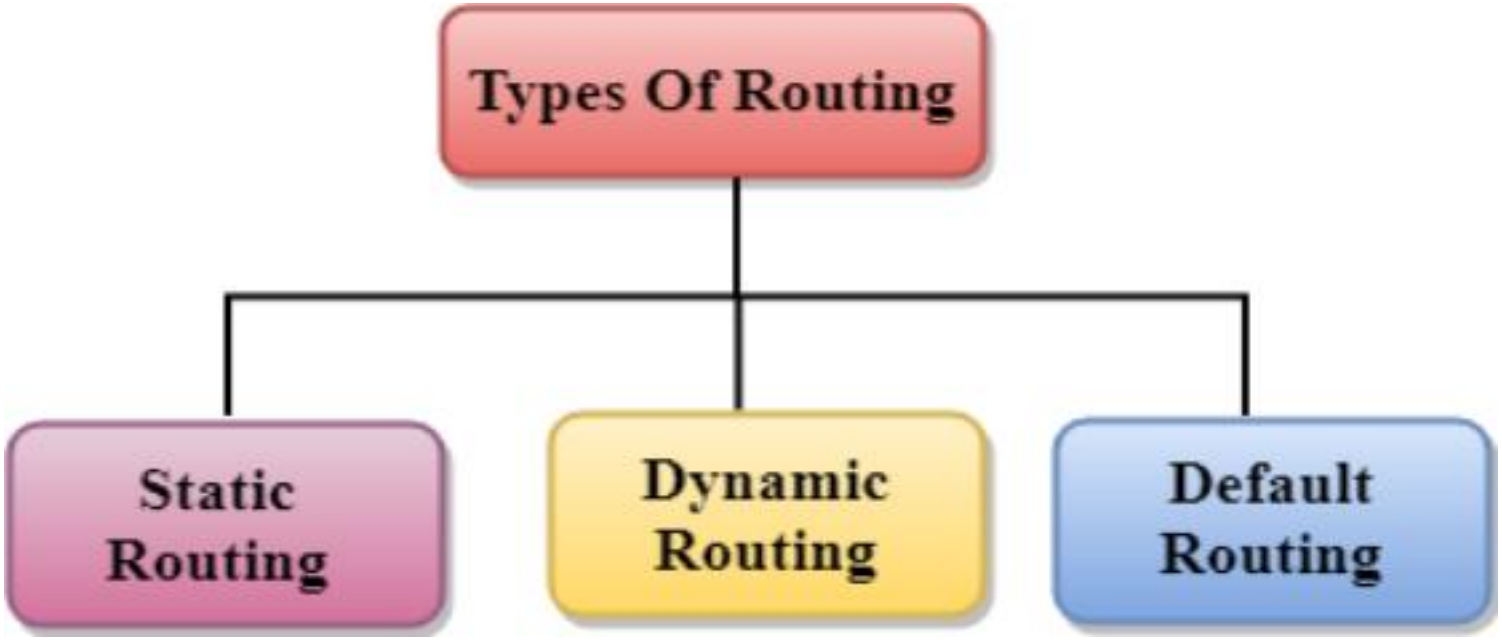
Prof .Vaishalee Joishar
CSE-Cyber Security
PIET,Parul University

Router

The router is a physical internetworking device that is designed to receive, analyze, and forward data packets between computer networks.



Connection of networks through Router


Types Of Routing
Static Routing
Dynamic Routing
Default Routing

Static Routing

In static routing, network administrators manually configure the routing tables on routers.

The routing paths do not change automatically; they remain fixed unless the administrator makes manual changes.

It is suitable for small networks with a stable topology.

Dynamic Routing

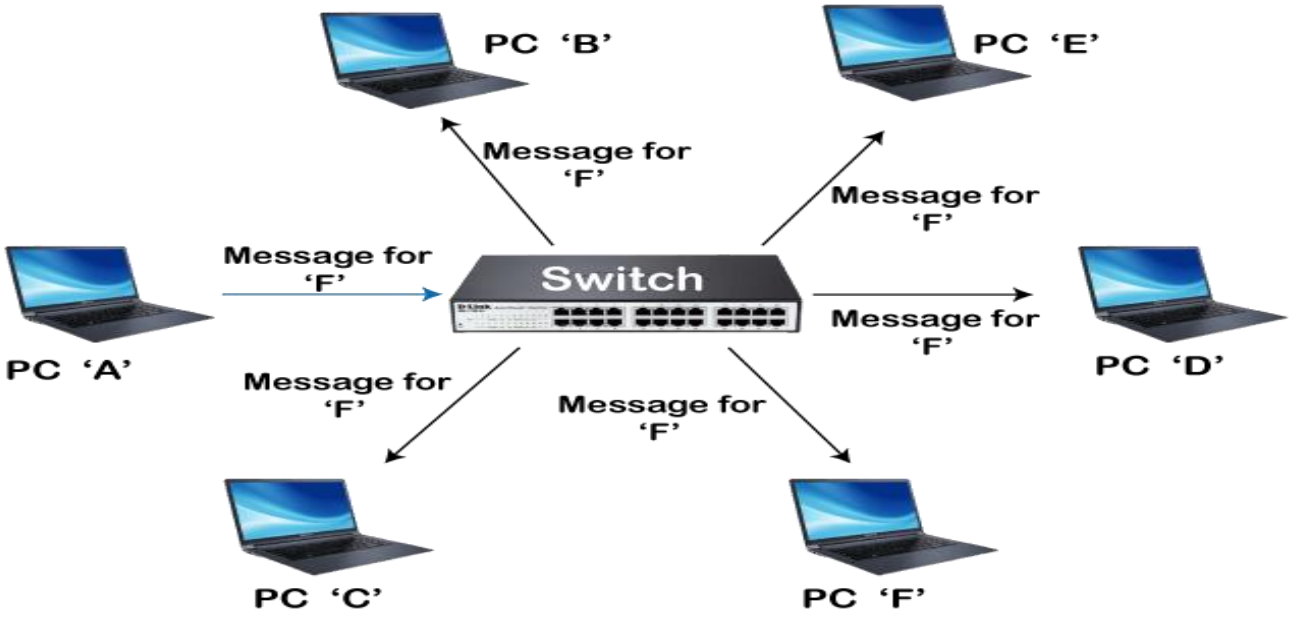
Dynamic routing protocols allow routers to automatically discover and communicate with each other to determine the best path for data packets.
Examples of dynamic routing protocols include RIP (Routing Information Protocol), OSPF (Open Shortest Path First), and EIGRP (Enhanced Interior Gateway Routing Protocol).
Dynamic routing is more scalable and adaptable to changes in network topology

Default Routing

Default routing is used when a router does not have specific information about the destination network in its routing table. The router forwards the packet to a default gateway, which then determines the next hop.

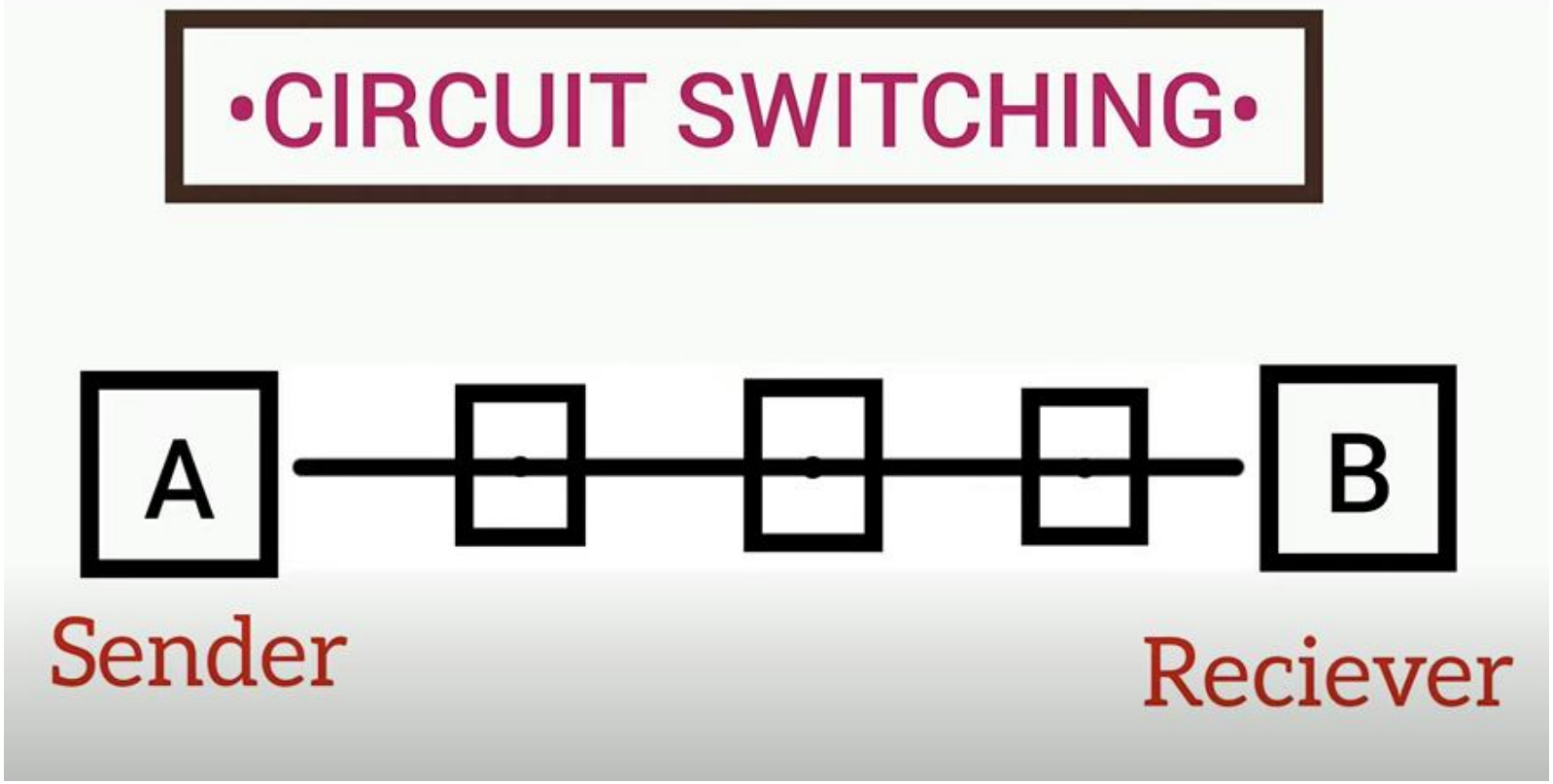
Switches

A network switch connects devices in a network to each other.



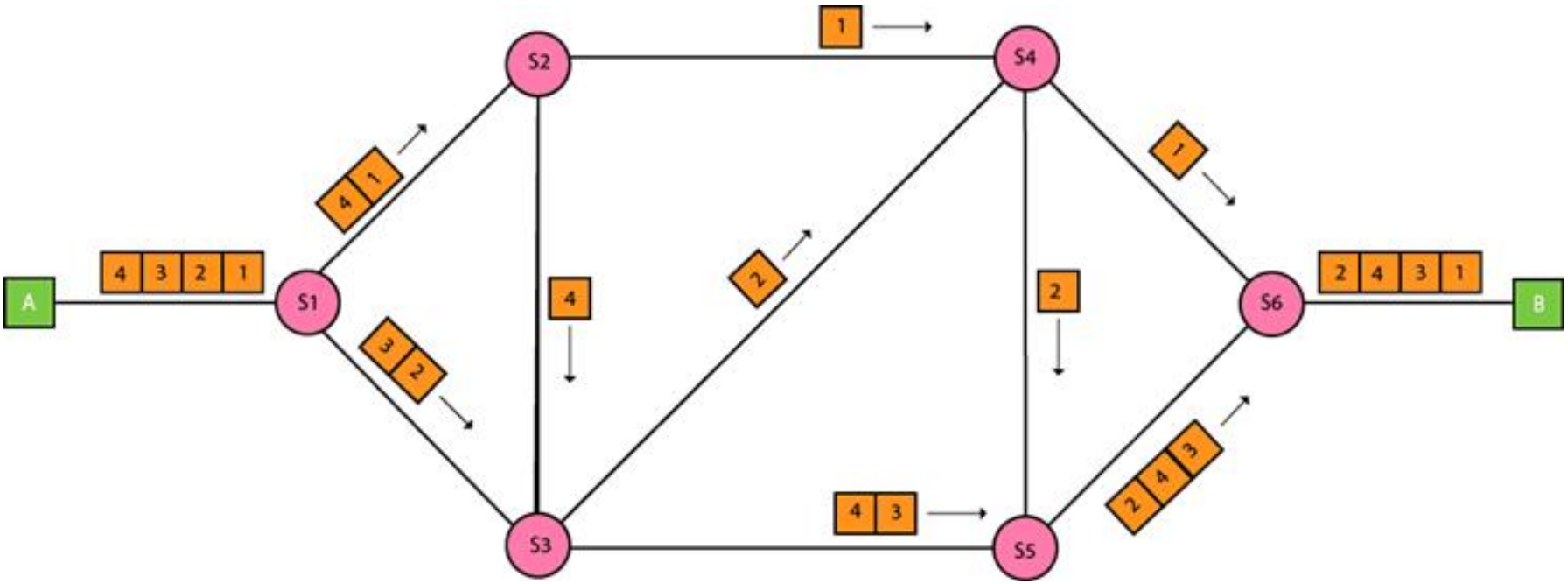
Circuit switching

Circuit switching is a method of communication in telecommunications networks where a dedicated communication path or circuit is established between two devices for the duration of their conversation. This is in contrast to packet switching, where data is divided into packets that are individually routed to their destination and reassembled.


•CIRCUIT SWITCHING•
A
B
Sender
Reciever

Packet switching

Packet switching is a method of communication in which data is broken down into small packets that are individually sent from the source to the destination over a shared network. This is in contrast to circuit switching, where a dedicated communication path is established for the entire duration of a conversation. Packet switching is the underlying technology of the Internet and many modern data networks.


A
4 3 2 1
S1
4 1
3 2
S2
4
S3
1
2
S4
4 3
S5
2
1
2 4 3
S6
2 4 3 1
B

Message switching

Message switching is an older and less common method of communication compared to circuit switching and packet switching. In message switching, complete messages or data units are sent from the source to the destination. Unlike packet switching, where data is divided into packets, and unlike circuit switching, where a dedicated path is established for the entire conversation, message switching involves forwarding entire messages.

•MESSAGE SWITCHING•

For example:- Email And Messages.

Endpoint solution

An "endpoint solution" typically refers to a comprehensive set of security tools and technologies designed to protect the various endpoints (devices) within a network. Endpoints include devices such as computers, laptops, servers, smartphones, tablets, and other devices that connect to a network. Endpoint security is crucial because these devices are often targeted by malicious actors seeking to exploit vulnerabilities and gain unauthorized access to sensitive data.

Reporting & Dashboard
Discovery
Spyware Blocking/ Removal
Inventory
Patch Management
Application Blocking
Correction Control Management
Network Access Control Add-On
Antivirus Add-On
Antivirus Auditing & Enforcement
Audit & Compliance Enforcement
Cloud Services Appliance
Host Intrusion Prevention
Endpoint Security Management

Types of Endpoint solutions:

Antivirus and Antimalware:
Function: Real-time scanning and protection against viruses, malware, and other malicious software.
Examples: Symantec, McAfee, Kaspersky.

Firewall Protection:
Function: Monitors and controls incoming and outgoing network traffic based on predefined security rules.
Examples: Windows Firewall, ZoneAlarm.

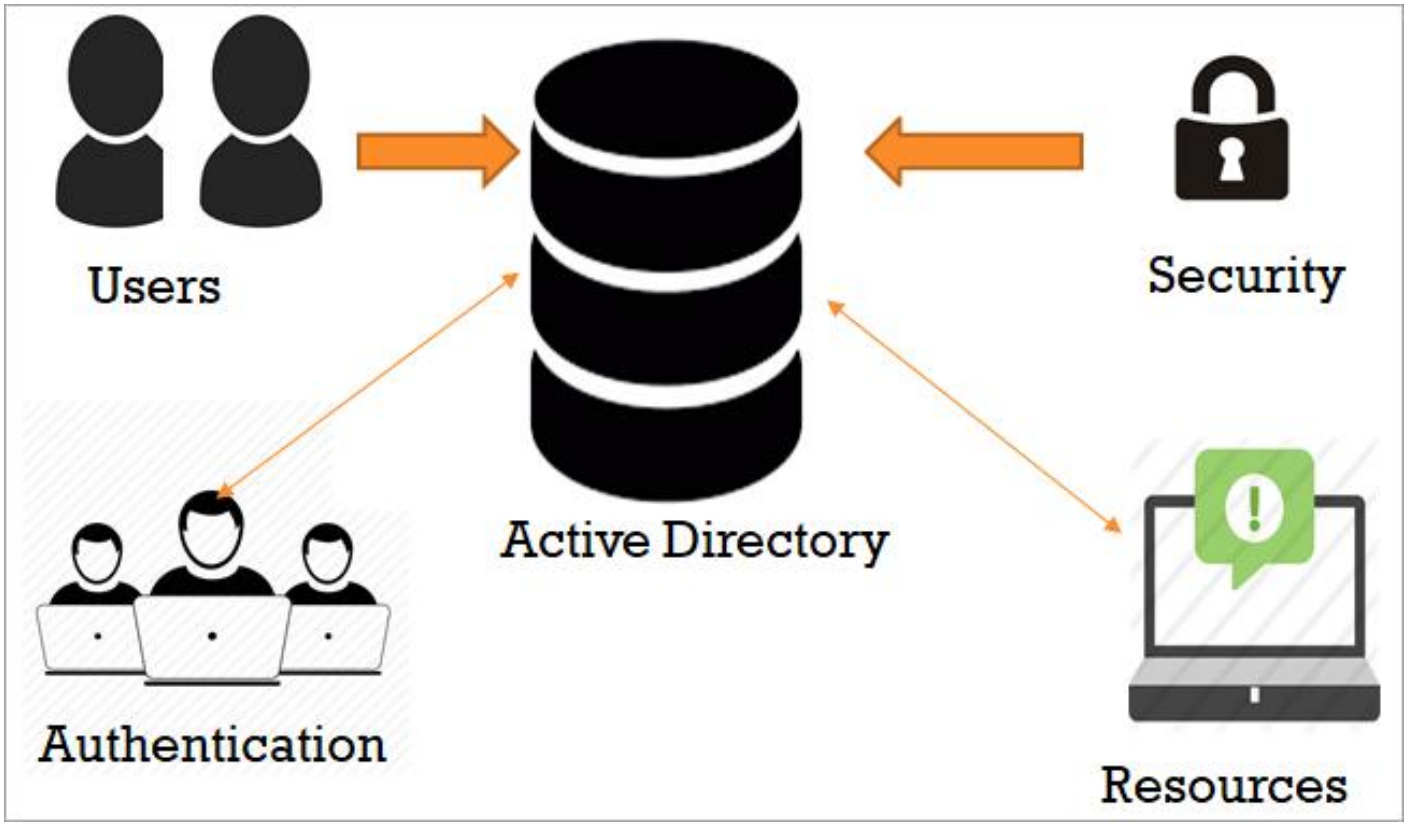
Endpoint Detection and Response (EDR):
Function: Monitors and responds to advanced threats and suspicious activities on endpoints in real-time.
Examples: CrowdStrike, Carbon Black, SentinelOne.

Device Control:
Function: Manages and restricts the use of removable storage devices, USB drives, and other peripherals.
Examples: Symantec Endpoint Protection, Trend Micro Apex One.

Access Directory

Directory service allows information to be stored, classified, and retrieved.
The directory service in Microsoft office is called Active Directory.

It is a database that is used to store an enormous amount of information about the User, User Groups, Client computers, and Network resources like Printers and Shared folders.


Users
Security
Active Directory
Authentication
Resources

Active Directory Structure

Active Directory structure, the top part is called a Forest. move down, it forms the tree which can be seen as a collection of domain and sub-domains.

The domain is at the center of the Windows Network.

Features Of Active Directory

It is a database of Objects. Some of these objects can be- Users, Groups, Printers, Devices, and computers.

It stores, organize and enables access to other objects.
It also provides essential networking services like DNS and Kerberos-based authentication.

If there is a network of hundreds or thousands of computers, an Active Directory is the only option to manage this network easily.

Active Directory governs the security policies when a VPN connection is established and people are allowed to connect to that network.

It allows Administrators to sit and centrally give users and computers permissions on the network and on individual computers.

The usage of Active Directory can vary from small organizations with a few hundred users to thousands of users across the world.

Tor Network

The Tor Network, short for The Onion Router, is a privacy-focused network that aims to enhance anonymity and security on the internet. It achieves this by directing internet traffic through a volunteer-operated network of servers, encrypting the data at each step in the process. The Tor Network is designed to protect users' privacy and anonymity by making it more challenging to trace their online activities and identity.

Features of the Tor Network:

Onion Routing:
The term "onion routing" refers to the layered encryption used by Tor. When data passes through the Tor network, it is encrypted in layers, much like the layers of an onion. Each server in the network decrypts one layer, revealing the instructions for where to send the data next. This process helps in obscuring the origin and destination of the data.

Volunteer-Operated Nodes:
The Tor Network relies on a decentralized network of volunteer-operated servers, known as nodes or relays.

Entry Nodes and Exit Nodes:
Entry nodes are the first nodes that a user's data passes through when entering the Tor network. Exit nodes are the final nodes through which data exits the Tor network and reaches its destination on the internet.

Anonymous Browsing:
Tor is commonly used to browse the internet anonymously. When a user accesses the internet through Tor, their IP address is hidden, making it more challenging for websites and online services to track their location and identity.

Hidden Services (.onion websites):
Tor allows the creation of hidden services, which are websites with addresses ending in ".onion." These sites can only be accessed through the Tor network, providing an additional layer of privacy for both the website and the user.

Encryption:
The data transmitted through the Tor network is encrypted, preventing eavesdroppers from understanding the content of the communication.

Circuits:
Tor creates circuits for each user, consisting of a series of nodes through which the user's data passes. Circuits are dynamically established and changed, enhancing security.

Privacy and Censorship Resistance:
Tor is often used by individuals seeking privacy in their online activities, as well as by those living in regions with internet censorship. It can help users bypass restrictions and access information that might be blocked by certain countries or organizations.

Networking Devices (Layer1,2,3)

Networking devices operate at different layers of the OSI (Open Systems Interconnection) model, which is a conceptual framework used to understand and design computer networks. Each layer in the OSI model serves specific functions, and various networking devices are designed to operate at different layers.

Examples of networking devices

Layer 1: Physical Layer
Hub:
Connects multiple network devices in a star topology at the physical layer. It operates by broadcasting data to all connected devices.

Repeater:
Extends the reach of a network by amplifying and retransmitting signals over long distances.

Layer 2: Data Link Layer

Switch:

Operates at the data link layer, forwarding frames based on MAC addresses.

Bridge:

Connects and filters traffic between different network segments based on MAC addresses.

Network Interface Card (NIC):
Hardware component that provides the interface between a device (e.g., computer) and the network, translating data between the device and the network.

Wireless Access Point (WAP):
Connects wireless devices to a wired network, allowing wireless communication based on the IEEE 802.11 standards.

Layer 3: Network Layer

Router:

Operates at the network layer, forwarding packets between different networks based on IP addresses.

Layer 3 Switch:

Combines features of a switch and a router, making routing decisions based on IP addresses. Layer 3 switches are often used to perform routing within a local network.

Multilayer Switch:

Similar to a layer 3 switch, a multilayer switch can operate at multiple layers of the OSI model, allowing for more complex networking functionality.

Network layer attacks

IP Spoofing:
In IP spoofing attacks, attackers forge the source IP address in the packets to make it appear as if the traffic is coming from a trusted source. This can be used to bypass access controls and launch various other attacks.

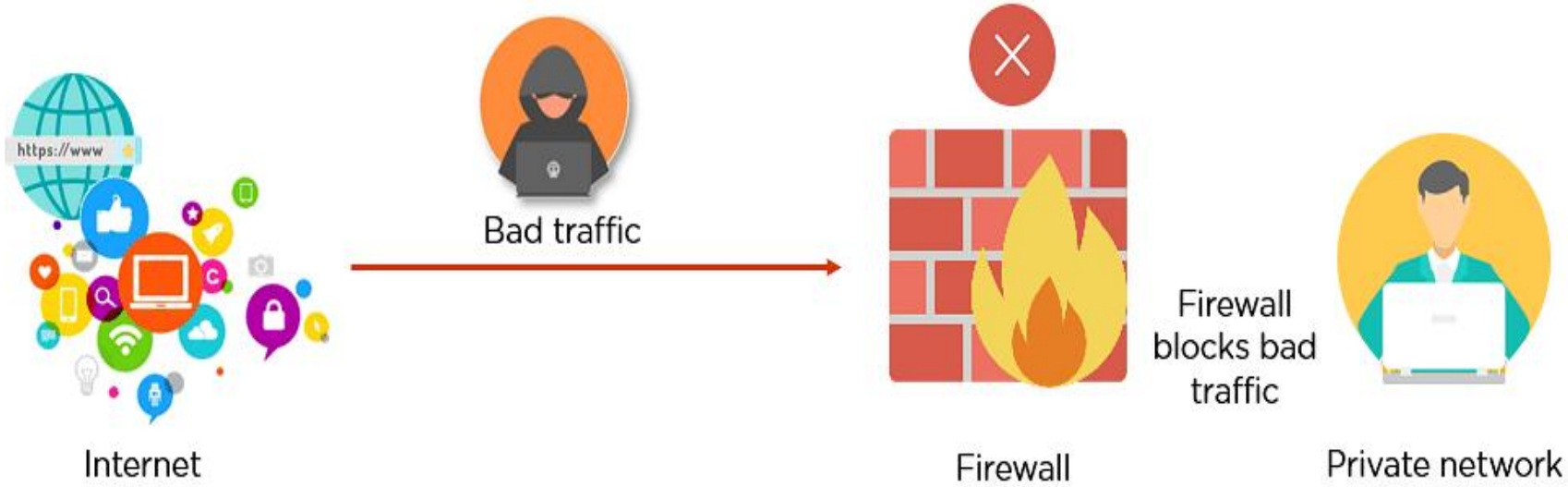
Denial of Service (DoS) Attacks:
DoS attacks flood a network, service, or website with traffic to overwhelm its resources and make it unavailable to legitimate users. Common forms include ICMP flood, SYN/ACK flood, and UDP flood attacks.

Session Hijacking:

Also known as session stealing, this attack involves an attacker intercepting and taking control of an established session between two parties.

Firewall

A firewall is a network security device or software that monitors and controls incoming and outgoing network traffic based on predetermined security rules. The primary goal of a firewall is to establish a barrier between a trusted internal network and untrusted external networks, such as the internet. Firewalls play a crucial role in protecting networked systems from unauthorized access, cyber attacks, and other security threats.


https://www
Bad traffic
Internet
Firewall
Firewall blocks bad traffic
Private network

Packet Filtering:
Firewalls examine individual data packets and filter them based on predetermined rules. Packet filtering involves making decisions about whether to allow or block packets based on factors like source and destination IP addresses, port numbers, and protocol types.

Proxy Services:
Proxy services involve an intermediary (proxy) between internal users and external servers. The proxy serves as a gateway, forwarding requests and responses between the internal network and the external network.

Stateful Inspection:

Stateful inspection, also known as dynamic packet filtering, involves tracking the state of active connections and making decisions based on the context of the traffic. This allows firewalls to understand the state of connections and make more informed decisions.

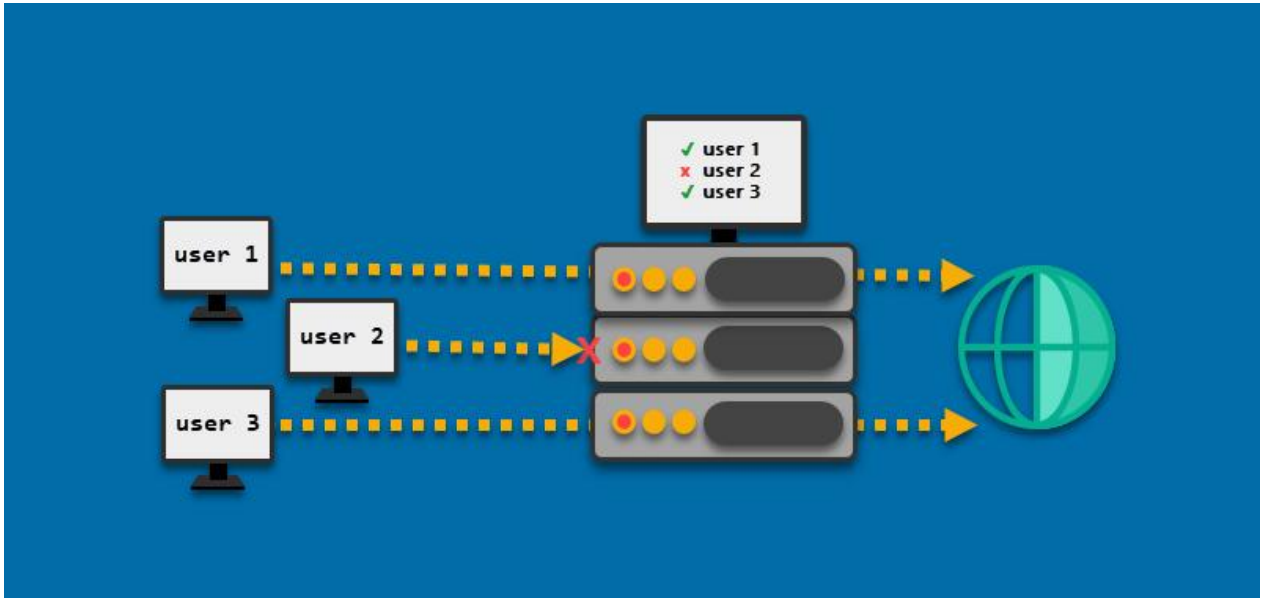
ACL(Access Control List)

An ACL works as a stateless firewall. While a stateful firewall examines the contents of network packets, a stateless firewall only checks if the packets follow the defined security rules.

How Does an ACL Work?

An ACL works as a stateless firewall. While a stateful firewall examines the contents of network packets, a stateless firewall only checks if the packets follow the defined security rules.

ACLs are tables containing access rules found on network interfaces such as routers and switches. When the user creates an ACL on a router or switch, the device becomes a traffic filter.


✓ user 1
x user 2
✓ user 3
user 1
user 2
user 3

Packet filtering

Packet filtering is a network security technique used to control the flow of network traffic based on predetermined criteria, such as source and destination IP addresses, port numbers, and protocols. This technique is implemented using devices like routers and firewalls that examine individual packets as they traverse the network and make decisions about whether to permit or deny their passage.

Key aspects of packet filtering

Filtering Criteria:
Packet filtering rules are defined based on various criteria, including:
Source and destination IP addresses: Filtering based on the source and destination addresses of the packets.

Port numbers: Filtering based on the port numbers associated with the communication.

Key aspects of packet filtering

Protocol type: Filtering based on the specific network protocol (e.g., TCP, UDP, ICMP).

Allow and Deny Decisions:
Packet filtering rules specify whether to allow or deny packets that match certain criteria. If a packet meets the conditions specified in an "allow" rule, it is permitted to pass through. Conversely, if it matches a "deny" rule, it is blocked.

Implicit Deny:
Packet filtering typically follows an "implicit deny" model, meaning that if a packet does not match any of the defined rules, it is implicitly denied by default. This ensures that administrators explicitly define the traffic they want to allow.

Stateless Inspection:
Packet filtering is stateless, meaning that each packet is evaluated independently of previous or subsequent packets. The device making filtering decisions does not maintain information about the state of active connections.

Router and Firewall Implementation:
Packet filtering is commonly implemented in routers and firewalls. Routers filter packets based on network layer criteria (e.g., source and destination IP addresses), while firewalls, especially stateful firewalls, can also filter based on transport layer criteria (e.g., port numbers).

Network Segmentation:
Packet filtering is often used to segment networks and control the flow of traffic between different network segments. This helps enforce security policies and restrict unauthorized access.

Access Control Lists (ACLs):
Access Control Lists (ACLs) are commonly used to implement packet filtering rules. ACLs define the criteria for filtering and specify the corresponding action (permit or deny).

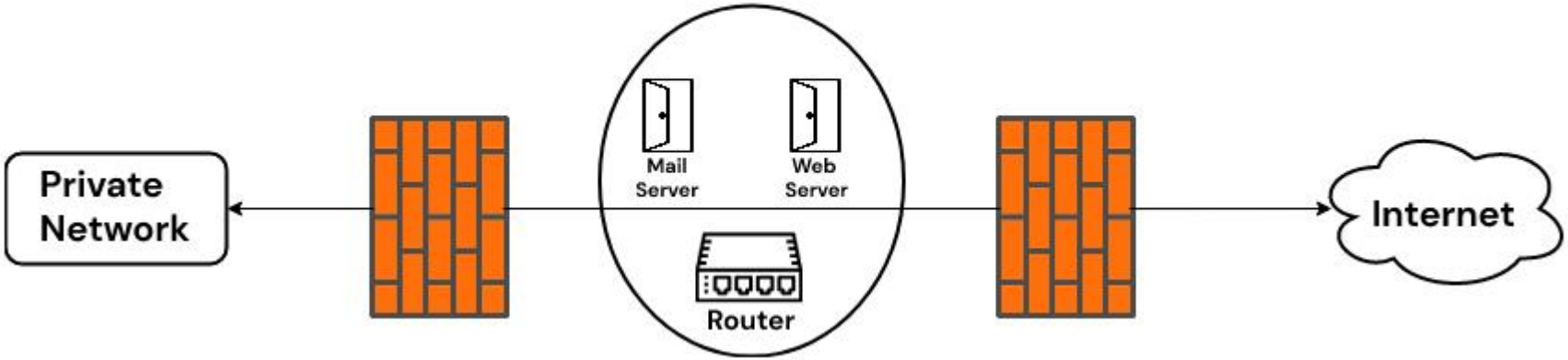
Performance Considerations:
Packet filtering is generally efficient in terms of performance because it operates at the network layer and can make decisions quickly based on header information in each packet.

DMZ(Demilitarized Zone)

A DMZ network provides a buffer between the internet and an organization's private network.

In computer security, a DMZ or demilitarized zone is a physical or logical subnetwork that contains and exposes an organization's external-facing services to an untrusted, usually larger, network such as the Internet.

Demiltarized Zone(DMZ)



Characteristics of a DMZ

Purpose:
The primary purpose of a DMZ is to host services that need to be accessible from external networks, such as the internet, while minimizing the potential risks to the internal network.

Isolation:
The DMZ is a neutral zone that acts as a buffer between the internal network and the external network. It isolates services and systems that are exposed to the internet, reducing the risk of direct attacks on internal assets.

Hosted Services:
Common services hosted in a DMZ include:
Web Servers: Hosting public-facing websites.
Mail Servers: Handling external email traffic.
FTP Servers: Allowing file transfers from external sources.
DNS Servers: Resolving domain names for external users.

Firewall Configuration:
Firewalls are used to control the traffic entering and leaving the DMZ. Typically, there are two firewalls:
External Firewall: Between the internet and the DMZ.
Internal Firewall: Between the DMZ and the internal network.

Security Measures:
Security measures within the DMZ may include intrusion detection and prevention systems, web application firewalls, and other security appliances to monitor and protect hosted services.

Proxy Servers and Reverse Proxies:
Proxy servers and reverse proxies may be used in the DMZ to enhance security. Proxies can filter and forward traffic, providing an additional layer of protection for internal systems.

Separation of Networks:
The DMZ allows for the separation of different network segments with varying levels of trust. External users can access services in the DMZ, but direct access to the internal network is restricted.

Security Policies:
Security policies are established to govern the access and usage of resources in the DMZ. These policies define the rules for interacting with external services and help maintain a secure network environment.

Monitoring and Logging:
The DMZ is typically monitored closely, and logs are regularly reviewed to identify and respond to any suspicious activities or security incidents.

Alerts and Audit Trails

Alerts and audit trails are essential components of network security and monitoring. They play a crucial role in detecting and responding to security incidents, ensuring compliance with security policies, and providing a detailed record of activities within a network.

Alerts:

Purpose:
Alerts are notifications generated by security systems or monitoring tools to inform administrators or security personnel about potential security incidents, anomalies, or predefined events.

Event Types:
Alerts can be triggered by various events, including:
Unusual network traffic patterns.
Security policy violations.
Suspicious login attempts.
Malware or intrusion detection.
Changes in system configurations.

Real-Time Notification:
Alerts are often configured to provide real-time notifications to administrators or a security operations center (SOC) when unusual or potentially malicious activities are detected.

Prioritization:
Alerts may be categorized and prioritized based on the severity of the detected event. High-priority alerts may require immediate attention, while lower-priority alerts may be reviewed periodically.

Integration with Incident Response:
Alerts are a critical component of incident response. They trigger the initiation of incident response procedures to investigate, contain, and mitigate security incidents.

Examples:
Intrusion detection system (IDS) alerts.
Firewall alerts for blocked or suspicious traffic.
Anomalies detected by network monitoring tools.

Audit Trails

Purpose:
Audit trails are chronological records of events and activities that occur within a network. They serve as a detailed log for auditing, compliance, and forensic analysis purposes.

Logged Events:
Events recorded in audit trails include:
User logins and logouts.
File or resource access.
Configuration changes.
System and application events.
Security policy changes.

User Accountability:
Audit trails help establish user accountability by recording the actions of users within the network. This is crucial for investigating security incidents and ensuring compliance.

Compliance Requirements:
Many industries and organizations are subject to regulatory compliance requirements that mandate the collection and retention of audit logs. Audit trails help demonstrate adherence to these regulations.

Forensic Analysis:

Audit trails play a vital role in forensic analysis by providing a detailed timeline of events. Security professionals can use audit logs to reconstruct the sequence of actions leading up to and following a security incident.

Monitoring and Analysis:
Security teams regularly monitor and analyze audit trails to identify unusual patterns, detect insider threats, and ensure that systems are being used in accordance with security policies.

Integrity and Non-Repudiation:
Audit trails contribute to data integrity and non-repudiation. They provide a record of who did what and when, making it difficult for users to deny their actions.

Examples:
Security Information and Event Management (SIEM) system logs.
Windows Event Logs.
Database audit logs.
Network device logs.

Parul® University | NAAC GRADE A++

https://paruluniversity.ac.in/